

Laser Capture Microdissection (Adapted from the Arcturus protocols)

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1. Remove slides to be used from -80°C freezer to dry ice. Place on a Kimwipe or paper towel to thaw for 30-60 seconds. From our experience, about 1-4 slides will require about 1 hour of LCM time. More time is required for compartments/structures that are small, less time is required for lower resolution (and/or more highly abundant compartments). A good rule of thumb is that no more than 4 slides be prepared at a time.
2. If sampling will be based on EGFP or EYFP expression, the signal can be enhanced by fixing the section for 5 minutes in 4% paraformaldehyde in phosphate buffered saline at room temperature. Rinse for 30 seconds in DEPC treated distilled water.
3. Whether paraformaldehyde fixed or not, place the slice in 75% ethanol for 30 seconds.
4. Place the slice in DEPC water for 30 seconds.
5. (Optional, depending on the criteria for capture) Stain the sections using Histogene (Arcturus). Place the slide on a Kimwipe and cover the sections for 20 seconds with about 200 microliters of Histogene.
6. Wash for 30 seconds in DEPC water.
7. Dehydrate by incubating for 30 seconds each in 75% ethanol, 95% ethanol and 100% ethanol.
8. Place in xylene for 5 minutes. Remove and air dry for 5 minutes.
9. Put the slides in a slide tray within an airtight container with a drying agent (DriRite).
10. Capture specimens onto LCM caps using the Arcturus Veritas LCM according to their protocols (www.arctur.com). In brief, the compartment of

interest is identified based on staining, expression of a fluorescent marker, or other criteria. A UV laser cuts around the specimen and an IR capture laser fuses (spot welds) the cut region of the PEN surface to the cap to capture the specimen away from the slide. The cap is stored on a microfuge tube in liquid nitrogen until used to isolate the RNA.